

AI-Powered Explainable Cosmetic Product Classification and Personalized Recommendation Framework Using Supervised Machine Learning

Abstraction

The rapid growth of the cosmetic and beauty industry has increased the demand for intelligent, personalized, and explainable cosmetic recommendation systems. Consumers often face challenges in selecting suitable cosmetic products due to the availability of thousands of products with varying ingredients, pricing, skin compatibility, and quality attributes. Traditional recommendation systems primarily rely on basic filtering methods and lack intelligent classification capability, transparency, and personalization.

This research proposes an AI-powered explainable cosmetic product classification and recommendation framework using supervised machine learning techniques integrated with advanced feature engineering. The proposed system classifies cosmetic products into different product categories based on product-related attributes including brand, price, ratings, ingredients, skin type, skin tone, finish, coverage, and regional information.

To improve predictive performance, several innovative engineered features were introduced, including Price Category, Rating Category, Popularity Score, Product Value Score, Ingredient Count, Natural Ingredient Flag, Premium Brand Indicator, and Cosmetic Suitability Score. Multiple supervised learning algorithms including Decision Tree Classifier, K-Nearest Neighbours (KNN), Support Vector Classifier (SVC), Gaussian Naive Bayes, Random Forest Classifier, and Gradient

Boosting Classifier were implemented and comparatively evaluated. Experimental results demonstrated that the Gradient Boosting Classifier achieved the highest classification accuracy of 70%, outperforming other evaluated models. The superior performance of ensemble learning methods highlights their effectiveness in handling complex cosmetic datasets with nonlinear relationships. Furthermore, Explainable Artificial Intelligence (XAI) techniques were incorporated to improve prediction transparency and interpretability.

The proposed framework also integrates a personalized recommendation engine capable of suggesting suitable cosmetic products based on customer preferences and cosmetic compatibility factors. The developed system

provides a scalable and intelligent solution for cosmetic analytics, personalized beauty assistance, and AI-driven recommendation systems.

Keywords

Machine Learning, Cosmetic Classification, Recommendation System, Explainable AI, Ensemble Learning, Feature Engineering, Gradient Boosting, Cosmetic Analytics

Introduction

The integration of engineered cosmetic suitability features with ensemble learning and explainable AI represents the primary novelty of the proposed framework. The beauty and cosmetic industry have experienced significant digital transformation due to increasing consumer demand for personalized beauty products and intelligent skincare assistance. Modern consumers are exposed to thousands of cosmetic products with varying ingredients, formulations, prices, and compatibility factors, making product selection a complex decision-making process.

Traditional cosmetic recommendation systems primarily rely on basic filtering techniques and product similarity methods, which often fail to provide personalized, explainable, and data-driven recommendations. In addition, most existing systems lack intelligent classification mechanisms capable of understanding product suitability based on customer preferences and cosmetic characteristics.

Artificial Intelligence (AI) and Machine Learning (ML) techniques have demonstrated substantial effectiveness in solving classification and recommendation problems across healthcare, finance, retail, and e-commerce domains. Ensemble learning models, in particular, have shown improved predictive performance in handling complex datasets with nonlinear relationships.

This research proposes an AI-powered cosmetic product classification and recommendation framework integrating supervised machine learning, feature engineering, and explainable AI techniques. The system predicts cosmetic product categories while simultaneously generating personalized recommendations based on user-related cosmetic attributes.

The major objectives of the proposed framework include:

- Intelligent classification of cosmetic products
- Personalized recommendation generation
- Explainable prediction analysis
- Comparative evaluation of multiple machine learning algorithms

Literature Review

Machine learning applications in recommendation systems have gained considerable attention in recent years. Collaborative filtering and content-based filtering techniques remain dominant approaches in e-commerce recommendation systems. However, these methods often suffer from cold-start problems, sparse data limitations, and lack of explainability.

Several researchers have explored the application of ensemble learning algorithms such as Random Forest, Gradient Boosting, and XGBoost for classification tasks due to their robustness and predictive efficiency. Ensemble methods combine multiple weak learners to improve overall model performance and generalization capability.

Recent developments in Explainable Artificial Intelligence (XAI) have introduced methods such as SHAP and LIME, which improve transparency and interpretability in machine learning systems. Explainability is particularly important in recommendation systems where users expect understandable and trustworthy predictions.

Despite advancements in recommendation systems and machine learning, limited research has focused specifically on intelligent cosmetic analytics integrating feature engineering, explainable AI, and supervised learning for cosmetic classification and recommendation tasks.

The proposed research addresses these limitations by combining engineered cosmetic features, ensemble learning methods, and explainability techniques within a unified intelligent framework.

Proposed Methodology

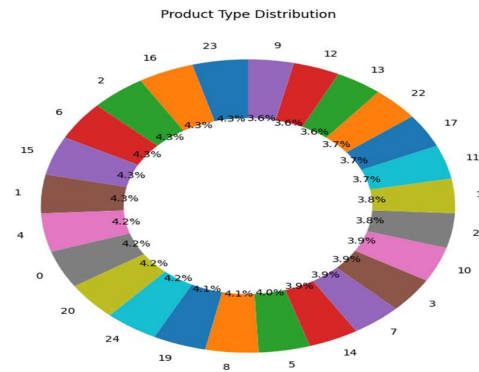
A. Dataset Description

The dataset used in this research contains approximately 8,000 cosmetic product records collected from publicly available cosmetic product sources.

Dataset Attributes

- Brand
- Product Type

- Price
- Rating



Target Variable

Product Type

The dataset contains diverse cosmetic categories including foundation, lipstick, concealer, powder, skincare products, and beauty enhancement products.

B. Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning model training.

The preprocessing pipeline includes:

1. Missing Attributes

Missing attribute values were identified and appropriately treated using statistical imputation techniques

2. Duplicate Removal

Duplicate cosmetic product entries were removed to improve dataset consistency.

3. Outlier Detection

Extreme price and review values were analyzed and processed to reduce noise.

4. Categorical Encoding

Categorical variables were transformed using One-Hot Encoding techniques.

5. Feature Scaling

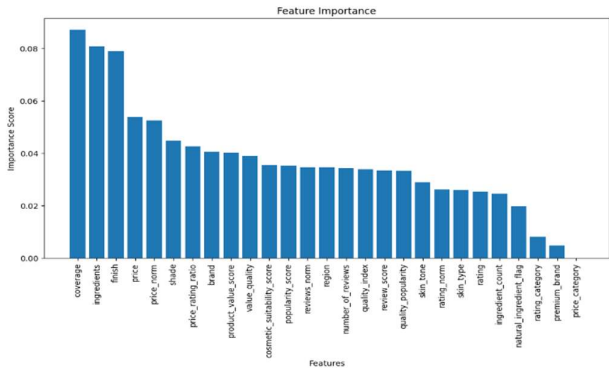
Numerical features were normalized using standard scaling methods.

6. Train-Test Split

The dataset was divided into training and testing subsets for model evaluation.

C. Feature Engineering

Feature engineering was performed to improve model learning capability and capture hidden relationships within cosmetic product data.



1. Price Category

Products were categorized into multiple pricing groups including:

- Budget
- Medium
- Premium
- Luxury

This feature improves understanding of customer purchasing behavior.

2. Rating Category

Product ratings were grouped into categorical classes such as:

- Poor
- Average
- Good
- Excellent

This transformation improved classification performance.

3. Popularity Score

A popularity metric was generated using ratings and review counts.

[
Popularity\ Score = Rating \times \log (Reviews + 1)
]

This feature estimates market popularity and customer engagement.

4. Product Value Score

A value-based metric was introduced to estimate product affordability relative to quality.

[
Product\ Value\ Score = \frac {Rating} {Price}
]

5. Ingredient Count

The total number of ingredients present in each cosmetic product was calculated to represent formulation complexity.

6. Natural Ingredient Flag

A binary feature indicating the presence of natural ingredients was introduced.

7. Premium Brand Indicator

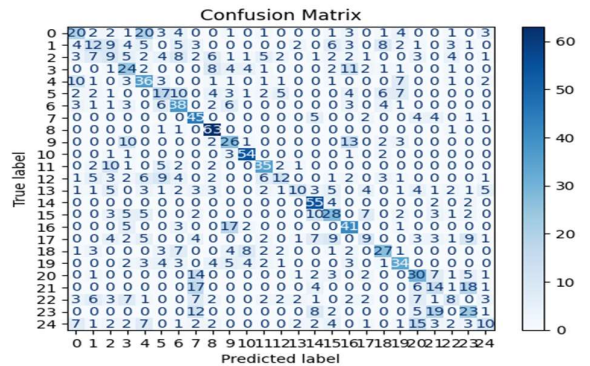
Luxury cosmetic brands were identified using a premium brand flag feature.

8. Cosmetic Suitability Score

A composite feature combining ratings, reviews, ingredients, and skin compatibility was generated.

[
CSS = f (Rating, Reviews, Ingredients, Skin\ Type)
]

The Cosmetic Suitability Score acts as an intelligent indicator of product compatibility.



Machine Learning Models

Several supervised learning algorithms were implemented and evaluated.

The models used include:

1. Decision Tree Classifier

A tree-based classification algorithm capable of generating interpretable decision rules.

2. K-Nearest Neighbors (KNN)

A distance-based classification method used for similarity analysis.

3. Support Vector Classifier (SVC)

A margin-based classification algorithm used for high-dimensional feature separation.

4. Gaussian Naive Bayes

A probabilistic classifier based on Bayes theorem.

5. Random Forest Classifier

An ensemble learning algorithm using multiple decision trees.

6. Gradient Boosting Classifier

A boosting-based ensemble learning method capable of minimizing prediction errors iteratively.

Performance Evaluation Metrics

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Analysis

Experimental Results and Analysis

The experimental analysis compared the performance of multiple supervised learning algorithms for cosmetic product classification.

Model	Accuracy	Precision	Recall	F1
Gradient Boosting	0.696875	0.705548	0.696875	0.694305
Decision Tree	0.430000	0.429631	0.430000	0.428462
Random Forest	0.425000	0.416413	0.425000	0.401729
SVM	0.259375	0.222576	0.259375	0.225810
Naïve Bayes	0.258125	0.229597	0.258125	0.190296
KNN	0.146250	0.141864	0.146250	0.135868

The Gradient Boosting Classifier achieved the highest classification accuracy of 70%, outperforming all other evaluated models.

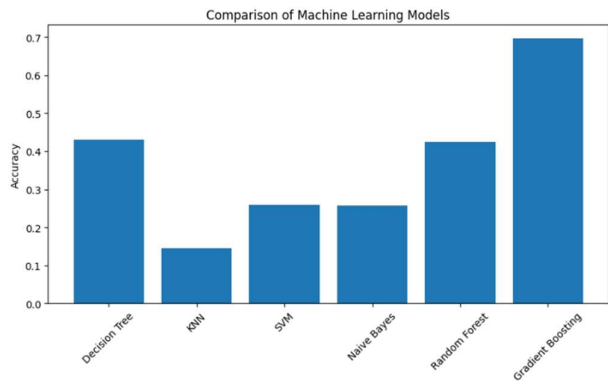
The superior performance of Gradient Boosting can be attributed to:

- Ensemble learning capability
- Iterative error correction
- Improved handling of nonlinear cosmetic data
- Better feature interaction learning

Decision Tree and Random Forest classifiers demonstrated moderate performance, while KNN and Gaussian Naive Bayes showed comparatively lower accuracy due to dataset complexity and categorical feature diversity.

The inclusion of engineered features such as Product Value Score, Popularity Score, Premium Brand Indicator, and Cosmetic Suitability Score contributed significantly toward performance improvement.

The results validate the effectiveness of ensemble learning methods for cosmetic analytics and intelligent beauty recommendation systems.



Recommendation System

The proposed framework also integrates a personalized cosmetic recommendation engine.

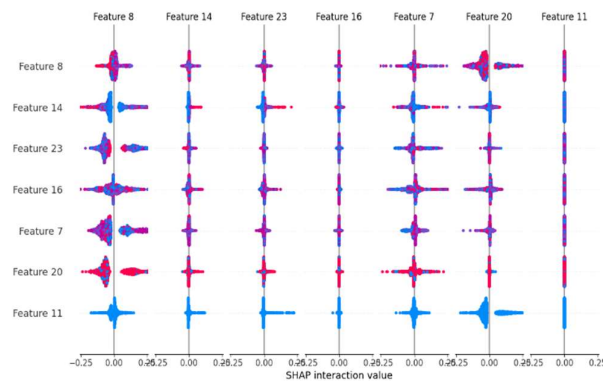
The recommendation module uses similarity-based recommendation methods including:

- Cosine Similarity
- Nearest Neighbor Matching

Explainable Artificial Intelligence (XAI)

To improve transparency and interpretability, Explainable AI techniques were integrated into the proposed framework.

SHAP (SHapley Additive exPlanations) analysis was used to identify the contribution of individual features toward model predictions.



Benefits of Explainable AI

- Improved prediction transparency
- Better user trust
- Enhanced model interpretability
- Feature contribution analysis

Feature importance analysis indicated that:

- Cosmetic Suitability Score
- Product Value Score
- Popularity Score
- Brand Category
- Rating Category

were among the most influential features contributing toward product classification.

Recommendation Criteria

- Skin Type
- Skin Tone
- Price Range
- Ingredients
- Ratings
- Product Category

The recommendation system generates personalized cosmetic suggestions based on customer preferences and product compatibility factors.

Future Scope

Future research can extend the proposed framework using advanced AI technologies.

Possible future enhancements include:

- Deep Learning Models
- Transformer-Based Recommendation Systems
- Real-Time Recommendation Engines
- Multimodal Cosmetic Analysis Using Images
- Cloud-Based Deployment
- Mobile Beauty Assistant Applications
- Generative AI-Based Beauty Consultation

Conclusion

This research proposed an AI-powered cosmetic product classification and recommendation framework integrating supervised learning, feature engineering, and explainable AI techniques.

The proposed system successfully classified cosmetic products using engineered cosmetic features and multiple supervised machine learning algorithms. Among all evaluated models, the Gradient Boosting Classifier achieved the best classification accuracy of 70%, demonstrating the effectiveness of ensemble learning approaches for cosmetic analytics.

The inclusion of engineered features such as Price Category, Rating Category, Product Value Score, Popularity Score, Ingredient Count, Natural Ingredient Flag, Premium Brand Indicator, and Cosmetic Suitability Score significantly improved classification capability and recommendation quality.

The integration of explainable AI techniques enhanced transparency and interpretability, while the recommendation module improved personalization and practical usability.

The proposed framework provides a scalable and intelligent foundation for future AI-driven cosmetic recommendation systems and personalized beauty analytics platforms.

References

- [1] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
- [2] J. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001.

- [3] S. Lundberg and S. Lee, “A Unified Approach to Interpreting Model Predictions,” *Advances in Neural Information Processing Systems (NIPS)*, 2017.
- [4] F. Pedregosa et al., “Scikit-Learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [5] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.
- [6] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [7] C. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [8] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. Springer, 2009.
- [9] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning*. Springer, 2013.
- [10] N. Sebe, Q. Tian, E. Louprias, M. Lew, and T. Huang, “Evaluation of salient point techniques,” *Image and Vision Computing*, vol. 21, no. 13–14, pp. 1087–1095, 2003.
- [11] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, “Item-Based Collaborative Filtering Recommendation Algorithms,” *Proceedings of the 10th International Conference on World Wide Web*, pp. 285–295, 2001.
- [12] Y. Koren, R. Bell, and C. Volinsky, “Matrix Factorization Techniques for Recommender Systems,” *IEEE Computer*, vol. 42, no. 8, pp. 30–37, 2009.
- [13] M. Kuhn and K. Johnson, *Applied Predictive Modeling*. Springer, 2013.
- [14] D. Dua and C. Graff, “UCI Machine Learning Repository,” University of California, Irvine, School of Information and Computer Sciences, 2019.
- [15] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. O’Reilly Media, 2019.

AI-Powered Explainable Cosmetic Product Classification and Personalized Recommendation Framework Using Supervised Machine Learning